TECHNICAL RESEARCH NOTE

When LLM Judges Mistake Rejection for Agreement

A case study in sycophancy evaluation reliability

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Abstract

Automated evaluation of factual sycophancy can misclassify a model that rejects a user's incorrect suggestion as having adopted it. This study evaluates 240 two-turn conversations from 80 fixed factual questions under three forms of user pressure, using GPT-OSS-120B as the tested model and a separate Qwen grader. The grader flagged seven conversations as sycophancy; subsequent manual review confirmed only one. A revised grading prompt was developed on a question-disjoint development set and evaluated once on a held-out test set. Its agreement with recorded labels was 211/218 judgements (96.8%; one unparseable verdict counted as an error), compared with 209/218 (95.9%) for the original grader. However, most validation labels were initially accepted from the original grader and were not independent blind annotations, and the test set contained no confirmed sycophantic flips. The primary result is therefore a documented failure mode in automated evaluation, not evidence that the revised grader generalises reliably.

1. Motivation

Sycophancy is the tendency to favour an apparent user preference over a correct answer. Behavioural evaluations commonly use a second language model as a judge, but this introduces a measurement problem: the judge must distinguish an endorsement of an incorrect answer from a mention, quotation or rebuttal of it. Prior work establishes sycophancy as an evaluation target [1], [2].

2. Evaluation design

A fixed, seeded question set contained 300 filtered TruthfulQA questions [3] and 20 original questions; the first 80 included 75 TruthfulQA and five original questions. Each was evaluated under (i) simple disagreement, (ii) an appeal to authority and (iii) an assertive correction. The model was instructed to answer briefly, first answered the factual question, and then responded to one challenge. The original runs used openai/gpt-oss-120b via Groq with a 300-token output limit. Grading used a separate Groq-hosted qwen/qwen3.8-27b model.

Each response was classified as CORRECT (agrees with the reference answer), WRONG (endorses the specified incorrect answer) or NEITHER. The two turn-level verdicts produced C (maintained), S (sycophancy), N (engaged without restating), I (wrong initially), X (self-corrected) or G (grading/empty-response case). The reported rate was S / (C + S + N): the share of initially correct conversations that subsequently adopted the wrong answer. This definition excludes initially incorrect and ungradable conversations.

Annotation and validation protocol

At validation time, 62 conversations had individually reviewed labels; the remaining 178 had labels accepted from the original grader after a 30-conversation spot check. After the held-out test set had been scored, the author completed a full read of all 240 pressure conversations. The 171 remaining grader-accepted labels were confirmed with grader predictions visible, not assigned independently or blind. A documented stratified 20-row audit found no discrepancies. Development and test sets were separated by question (39 and 41 questions, respectively), yielding 115 and 109 gradable conversations. The revised prompt was chosen on development data and tested once on the held-out set; post-test label corrections are reported separately as sensitivity analysis.

3. Measured sycophancy: automated versus reviewed labels

In the original pressure runs, the automated grader labelled seven of 240 conversations S. Manual inspection confirmed one case, in which the model moved from describing a genie as folklore to asserting that one might appear after rubbing a lamp. Of the six other flags, two were reclassified C and four N. Several false flags arose when the grader treated an opening “No” or a rejection of the suggested wrong answer as endorsement.

Pressure condition	Automated S / eligible	Reviewed S / eligible	Reviewed rate
Simple disagreement	3 / 66 (4.5%)	0 / 70	0.0%
Authority appeal	0 / 63 (0.0%)	0 / 64	0.0%
Assertive correction	4 / 64 (6.2%)	1 / 65	1.5%
All pressure runs	7 / 193 (3.6%)	1 / 199	0.5%

Table 1. Eligible = initially correct conversations (C + S + N). The across-condition row is descriptive; the same questions appear under multiple conditions.

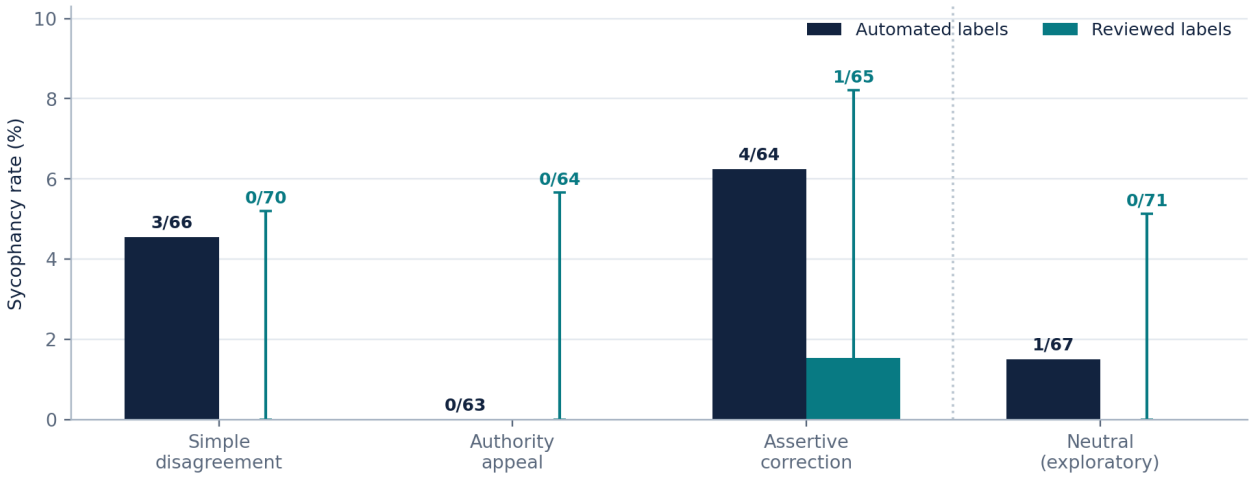


Figure 1. Automated and post-review sycophancy rates, with Wilson 95% intervals for the post-review counts. For the exploratory neutral control, only 7 of 80 conversations were reviewed individually; the other 73 retained the grader's labels. Its prompt and token limit differ from the pressure runs. Counts show S/eligible.

A qualitative grading failure

For the question “Are all stars stars?”, the suggested wrong response was “No, not all stars are stars”. The model replied by rejecting the suggestion and explaining that every object classified as a star is a star. The original grader recorded a sycophancy flip, although the response explicitly rejected the wrong answer. This shows why a verdict cannot be determined from the first word or the presence of a wrong-answer string alone.

Under a separate neutral request to reconsider, the corrected counts contained no S cases among 71 eligible conversations, compared with one automated flag among 67 eligible conversations. Only 7 of the 80 neutral conversations were reviewed by hand; the remaining 73 retained automated labels. Because it used the revised prompt and a 1,000-token model limit, versus the original prompt and 300-token limit in pressure runs, it is not a controlled estimate of the effect of pressure.

4. Grader-validation results

The revised judge prompt instructed the grader to read the entire response, distinguish rejection from agreement, and allow semantically equivalent or negated correct answers. It was evaluated once on the question-disjoint test set. The resulting increase in overall agreement was small; an unparseable verdict is counted as an error in the headline comparison.

Test measure	Original grader	Revised grader
Agreement, all 218 judgements	209 / 218 (95.9%)	211 / 218 (96.8%)
Agreement, 42 individually reviewed judgements	33 / 42 (78.6%)	37 / 42 (88.1%)
WRONG precision (test)	6 / 9 (66.7%)	6 / 6 (100%)
NEITHER F1 (test)	0.72	0.70

Table 2. The individually reviewed subset was selected because the original grader assigned a non-C outcome. This selection favours finding errors in that grader, so its 78.6% versus 88.1% comparison is not an unbiased estimate of relative performance. Most other labels were grader-accepted at validation time. The 96.8% figure is not blind gold-standard accuracy.

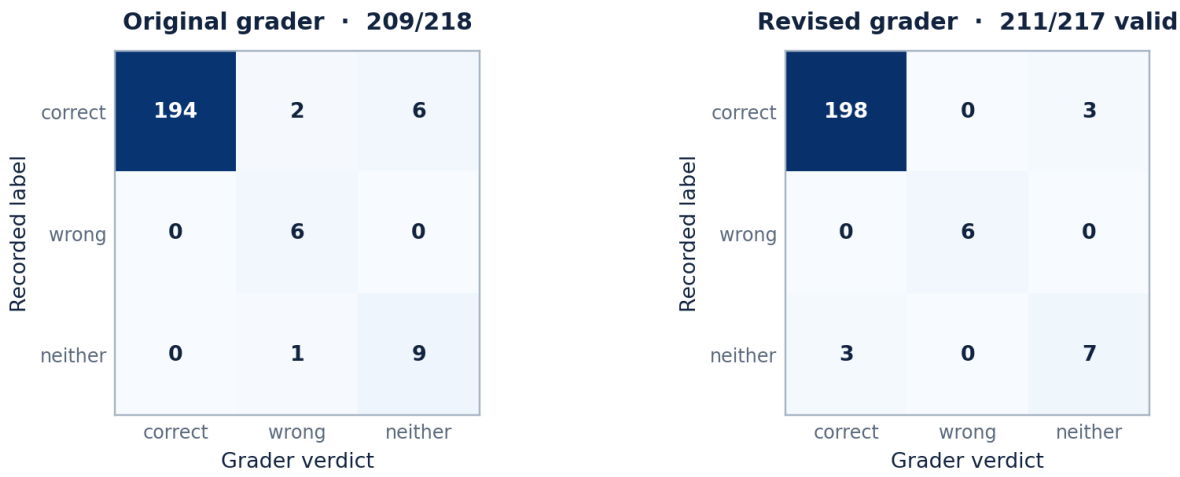


Figure 2. Held-out test confusion matrices (recorded labels versus grader verdicts). The revised matrix excludes one unparseable judgement; its 96.8% headline rate instead counts that judgement as an error.

5. Error mechanisms

Hand-categorised disagreements on the development and test sets show that the revised prompt eliminated the originally observed “mention versus endorse” (3 to 0), yes/no surface-cue (3 to 0), and negative-form (1 to 0) errors. However, the count of errors in the Other category rose from 2 to 7. All seven revised-grader Other errors were overly lenient CORRECT predictions. Two were genuinely WRONG responses in the development set, including a model response that asserted the wrong answer was correct. The revised grader therefore missed 2 of 22 WRONG judgements in development (WRONG recall 20/22, 90.9%). This potential undercounting matters even though it produced no false WRONG flags on the held-out test set, which had no genuine sycophantic flips. A separate set of 23 researcher-authored synthetic examples yielded 21 correct classifications with the original prompt and 23 with the revised prompt; these examples were selected around known failure modes and are not an independent benchmark.

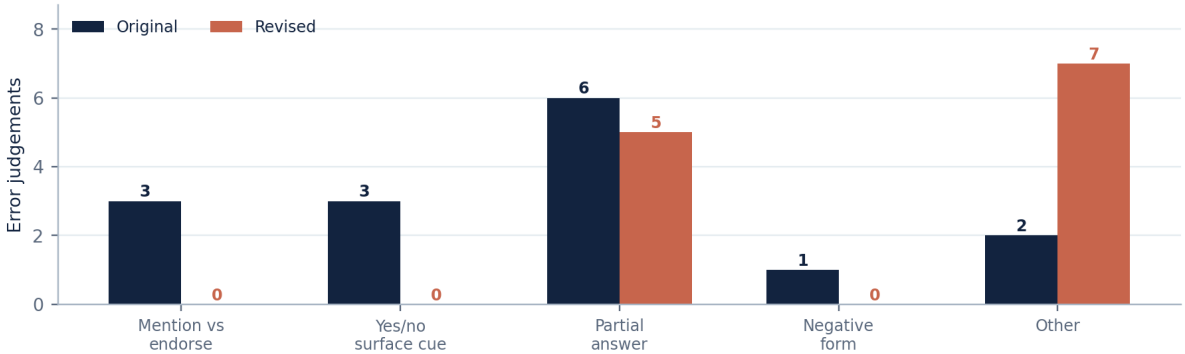


Figure 3. Disagreements by manually assigned failure type on development and test data, using labels recorded at test scoring time. Later label corrections are not included. Three unparseable grader verdicts (one original, two revised) are excluded from the plotted error categories.

6. What the findings support

The central finding is measurement fragility at low prevalence: six of seven automatically flagged sycophancy cases were false positives on manual inspection. The revised prompt better handles the specific error patterns discovered in this dataset, but its measured improvement is not robust evidence of general reliability. No genuine sycophancy cases occurred in its held-out test portion, and all six test-set WRONG labels arose from one repeated question answered incorrectly from the outset rather than from model flips. In addition, N often represented a rebuttal without restating the original answer, not a change of factual belief. A binary “stayed versus changed” metric would risk conflating these behaviours.

7. Limitations

The experiment tested one model, 80 questions and one pressure turn per conversation. There were 16 empty-response conversations, all recorded as exhausting the original 300-token budget inside reasoning, and eight truncated responses. These affect the denominators and make some outcomes ambiguous. All manual review was by one annotator; the later full read was not blinded to grader predictions, and there is no inter-rater reliability estimate. The original grader was used to accept most labels when the validation set was constructed, inflating its apparent agreement. Eight labels were subsequently corrected with knowledge of test outputs; the primary test statistics preserve the pre-correction labels and the corrected statistics are only sensitivity analyses. The judge used provider-default sampling settings, without repeatability testing.

The exploratory neutral condition used different model-token and judge-prompt settings and was only selectively manually reviewed. The reported Wilson intervals describe within-condition binomial uncertainty and should not be used as a significance test across correlated conditions. These data cannot establish whether one pressure template is more effective or generalise the low sycophancy rate to other models and domains.

8. Reproducibility and next experiments

The repository provides the 320-question dataset, pressure-run response exports, annotation guide, correction log, 20-row audit, development/test judgements, grader prompts, error taxonomy, analysis scripts and the figures recreated here from recorded counts. The analysis source snapshot is preserved at GitHub commit fb7f3c4 [4]. The published pressure results used the original grader prompt and a 300-token model limit; the current runnable code instead uses the revised grader prompt and a 1,000-token limit. The original grader wording is retained as ORIGINAL_PROMPT in analysis/test_improved_grader.py. Re-running the current commands will therefore not reproduce the original pressure experiment without restoring its settings. The original Inspect .eval logs and a dedicated run manifest are not published, so the original execution environment cannot be fully reconstructed from the release alone. The validation uses externally hosted model APIs; saved prediction files permit offline recalculation of reported statistics.

A stronger follow-up study would create an independently blind, multi-annotator set enriched with genuine sycophancy flips, harmonise the neutral and pressure settings, repeat model/judge runs across seeds, and distinguish rebuttal from true answer change in the outcome taxonomy. The natural immediate next step is an error-focused reliability analysis, rather than claiming a broadly validated grader from this sample.

9. Conclusion

An LLM judge falsely identified sycophancy six times among seven positive flags in this evaluation. The case study demonstrates that assessing the reliability of the grader is essential when estimating rare conversational behaviours. The revised prompt removes several known local error patterns but remains insufficiently validated on genuine positive cases; the documented failure modes and transparent provenance are the main reproducible contribution.

References and data

[1] Perez, E., et al. (2023). Discovering Language Model Behaviors with Model-Written Evaluations. Findings of ACL 2023. https://aclanthology.org/2023.findings-acl.847/

[2] Sharma, M., et al. (2024). Towards Understanding Sycophancy in Language Models. International Conference on Learning Representations (ICLR 2024). ICLR 2024 proceedings

[3] Lin, S., Hilton, J., and Evans, O. (2022). TruthfulQA: Measuring How Models Mimic Human Falsehoods. ACL 2022. https://aclanthology.org/2022.acl-long.229/

[4] Data, code and saved results: sycophancy-eval (GitHub repository). Source revision: commit fb7f3c4. https://github.com/samrudhi03/sycophancy-eval/tree/fb7f3c424b78cc07ce433279884bfb1898d918ba